
CURRICULUM VITAE

Lisa C. McManus

Hawai'i Institute of Marine Biology
School of Ocean and Earth Science and Technology
University of Hawai'i at Mānoa • 46-007 Lilipuna Road, Kāne'ohe, HI 96744
mcmmanusl@hawaii.edu • <https://lmcmanus47.github.io/>

ACADEMIC POSITIONS

Hawai'i Institute of Marine Biology, University of Hawai'i at Mānoa <i>Associate Research Professor</i>	Kāne'ohe, HI 2025-present
Hawai'i Institute of Marine Biology, University of Hawai'i at Mānoa <i>Assistant Research Professor</i>	Kāne'ohe, HI 2020-2025
Department of Ecology, Evolution, and Natural Resources Rutgers University <i>Postdoctoral Researcher</i>	New Brunswick, NJ 2017-2020

EDUCATION

Princeton University, Ph.D. in Ecology and Evolutionary Biology - NMFS-Sea Grant Population and Ecosystem Dynamics Fellow - National Defense Science and Engineering Graduate Fellow	Princeton, NJ 2012-2017
University of Miami, B.S., Marine and Atmospheric Science - Summa cum laude - Phi Beta Kappa	Miami, FL 2006-2010

I. RESEARCH

PEER- REVIEWED PUBLICATIONS

* Denotes equal author contribution

† Denotes student/postdoctoral/technician publication from the McManus Marine Ecological Theory Lab

‡ Denotes student/postdoctoral/technician publication from courses or workshops led by McManus

Manuscripts under review

Forrest DL*, **McManus LC***, Tekwa EW, Schindler DE, Colton MA, Webster MM, Fox HE, Essington TE, Palumbi SR, Mumby PJ, DeFilippo L, Schill SR, Pollock FJ & Pinsky ML (In review). Marine spatial planning to enhance coral adaptive potential. *Journal of Applied Ecology*.
bioRxiv preprint: <https://www.biorxiv.org/content/10.1101/2024.08.27.609972v2>

Vogt-Vincent NS[†], Rocha de Souza MS[†], Feloy K, Toonen RJ & **McManus LC** (In review). Growth rate and

adaptive capacity, not just thermal tolerance, are critical for long-term coral persistence under climate change. *Proceedings of the Royal Society B: Biological Sciences*.
bioRxiv preprint: <https://www.biorxiv.org/content/10.1101/2025.09.24.678233.full.pdf>

Buckner JH, Meunier ZD, Arroyo-Esquivel J, Fitzpatrick N[†], Greiner A, **McManus LC** & Watson JR (In review). Recovering complex ecological dynamics from time series using state-space universal dynamic equations.

Quinlan ZA, **McManus LC** & Donahue MJ (In review). Coral larval settlement preference for crustose coralline algae can enhance coral persistence and restoration success. *Ecological Applications*.

Gibbs TL[‡], Dahlin KJM[‡], Brennan J[‡], Silveira CB^{*} & **McManus LC**^{*} (In review). Coexistence of bacteria with a competition-colonization tradeoff on a dynamic coral host. *Theoretical Ecology*.
bioRxiv preprint: <https://www.biorxiv.org/content/10.1101/2024.09.15.612558v1>

Peer-reviewed articles

25. Rahnke SR[†], Winter KB, Tuttle Raz LJ & **McManus LC** (In press). Interaction strength and harvest intensity mediate predator-prey dynamics on coral reefs. *Ecosphere*.

24. Innes-Gold AA[†], Feloy K, Callan CK, Alegado RA, Ching C, Kawelo H, Kotubetey K, Lima B, Madin EMP, ME*E Lab, Rii YM, Winter KB & **McManus LC** (2025). Restoration and management of an Indigenous aquaculture system helps mitigate climate change impacts to estuarine fisheries. *npj Ocean Sustainability* 4(51). <https://doi.org/10.1038/s44183-025-00152-3>

23. Ferretti P^{*‡}, Martignoni MM^{*‡}, McManus LC^{*}, Sakal T[‡], Liaghat A[‡], Stevens B[‡], Dahlin KJM[‡], Souza LS[‡], Cardon ZG, Silveira CB, Bordenstein SR & Roughgarden J (2025). Theory of host-microbe symbioses: Challenges and opportunities. *Cell host & microbe* 33(7): 1052-1056.
<https://doi.org/10.1016/j.chom.2025.05.001>

22. Grottoli AG, Hulver AM, Vega Thurber R, Toonen RJ, Schmeltzer ER, Kuffner IB, Barott KL, Baums IB, Castillo KD, Chapron L, Coffroth MA, Combosch DJ, Correa AMS, Crandall ED, Donahue M, Eirin-Lopez JM, Felis T, Ferrier-Pagès C, Harrison HB, Heron SF, Huang D, Humanes A, Kenkel CD, Krueger T, Madin J, Matz MV, **McManus LC**, Medina M, Muller EM, Padilla-Gamiño JL, Putnam HM, Sawall Y, Shlesinger T, Sweet MJ, Voolstra CR, Weis VM, Wild C & Wu HC (2025). Future of coral bleaching research. *BioScience* 75(7): 585-598. <https://doi.org/10.1093/biosci/biaf066>

21. Lee E[†] & **McManus LC** (2025). Rate of temperature increase and genetic diversity drives marine metapopulation persistence. *The American Naturalist* 206(4): 318-334.
<https://doi.org/10.1086/737022>

20. Vogt-Vincent NS[†], Pringle JM, Cornwall CE & **McManus LC** (2025). Anthropogenic climate change will likely outpace coral range expansion. *Science Advances* 11(23): eadr2545.
<https://doi.org/10.1126/sciadv.adr2545>

19. Ong TW^{*}, **McManus LC**^{*}, Vasconcelos VV^{*}, Yang L & Su C (2025). Seeing halos: Spatial and consumer-resource constraints to landscape of fear patterns. *The American Naturalist* 205(6): 590-603.
<https://doi.org/10.1086/735688>

18. Chen GK[†], [McManus LC](#), Fan TY & Madin JS (2025). Relative contributions of size and shape to coral demography. *The American Naturalist* 205(6): 604-616. <https://doi.org/10.1086/735482>
17. Innes-Gold AA^{††}, Rahnke SA^{††} & [McManus LC](#) (2025). Land-sea interactions: Nutrient inputs, fishing effort, and predation shape estuarine fisheries harvest. *Estuarine, Coastal and Shelf Science* 323(30): 109377. <https://doi.org/10.1016/j.ecss.2025.109377>
16. Innes-Gold A[†], [McManus LC](#), Lester E, Ong TW, Cook McNab A, Rahnke S[†], Brett Pablo J[†], Tokoyoda A, Watson D & Madin EMP (2025). Herbivory and temperature mediate coral reef halo dynamics. *The American Naturalist*. <https://doi.org/10.1086/738015>
15. Amaral PAK, Salvador RD & [McManus LC](#) (2024). Media Arts on a Marine Research Station: Reflections on a Storytelling Course in Support of Native Hawaiian Communities. *Science Education and Civic Engagement* 16(2): 45-51.
14. Innes-Gold AA[†], Carvalho P, [McManus LC](#), Correa-Garcia S, Marcoux SD, Oleson KLL, Stokes K & Madin EMP (2024). Modeling the interactive effects of sea surface temperature, fishing effort, and spatial closures on reef fish populations. *Theoretical Ecology* 17(4): 289-300. <https://doi.org/10.1007/s12080-024-00591-7>
13. Asbury M^{††}, Innes-Gold AA^{†††}, Wulstein DM^{††}, Madin EMP, Madin JS & [McManus LC](#) (2024). Recovery potential of fish and coral populations following ecological disturbance. *Ecosphere* 15(6): e4915. <https://doi.org/10.1002/ecs2.4915>
12. Tilman AR, Krueger EH, [McManus LC](#) & Watson JR (2024). Maintaining human wellbeing as socio-environmental systems undergo regime shifts. *Ecological Economics* 221, 108194. <https://doi.org/10.1016/j.ecolecon.2024.108194>
11. Innes-Gold AA[†], Madin EMP, Stokes K, Ching C, Kawelo H, Kotubetey K, Reppun F, Rii Y, Winter KB and [McManus LC](#) (2024). Restoration of an Indigenous aquaculture system can increase reef fish density and fisheries harvest in Hawai'i. *Ecosphere* 15(3), e4797. <https://doi.org/10.1002/ecs2.4797>
10. McClanahan TR, Darling ES, Beger M, Fox HE, Grantham HS, Jupiter SD, Logan CA, Mcleod E, [McManus LC](#), Oddenyo RM, Suyra GS, Wenger AS & Zinke J (2024). Diversification of refugia types needed to secure the future of coral reefs to climate change. *Conservation Biology* 38(1), e14108. <https://doi.org/10.1111/cobi.14108>
9. Colton MA, [McManus LC](#), Schindler DE, Mumby PJ, Palumbi SR, Webster MM, Essington TE, Fox HW, Forrest DL, Schill SR, Pollock FJ, DeFilippo LB, Tekwa EW, Walsworth TE & Pinsky ML (2022). Conservation in a warming world must harness evolution. *Nature Ecology and Evolution* 6: 1405-1407. <https://doi.org/10.1038/s41559-022-01854-4>
8. DeFilippo LB^{*}, [McManus LC](#)^{*}, Pinsky ML, Colton MA, Webster MS, Essington TE, Palumbi SR, Mumby PJ & Schindler DE (2022). Assessing the potential for demographic restoration and assisted

evolution to build climate resilience in coral reefs. *Ecological Applications* 32: e2650.
<https://doi.org/10.1002/eap.2650>

7. van Woesik R, Shlesinger T, Grottoli AG, Toonen RJ, Vega Thurber R, Warner M, Hulver AM, Chapron, L, McLachlan RH, Albright R, Crandall E, DeCarlo TM, Donovan MK, Eirin-Lopez J, Harrison HB, Heron SF, Huang D, Humanes A, Krueger T, Madin JS, Manzello D, **McManus LC**, Matz M, Muller EM, Rodriguez-Lanetty M, Vega-Rodriguez M, Voolstra CR & Zaneveld J (2022). Coral-bleaching responses to climate change across biological scales. *Global Change Biology* 28: 4229-4250.
<https://doi.org/10.1111/gcb.16192>
6. **McManus LC***, Forrest DL*, Tekwa E, Schindler DE, Walsworth TE, Colton MA, Webster MS, Essington TE, Palumbi SR, Mumby PJ & Pinsky ML (2021). Evolution and connectivity influence the persistence and recovery of coral reefs under climate change in the Caribbean, Southwest Pacific, and Coral Triangle. *Global Change Biology* 27(18): 4307-4321.
<https://doi.org/10.1111/gcb.15725>
5. **McManus LC**, Tekwa E, Schindler DE, Walsworth TE, Colton MA, Webster MS, Essington TE, Palumbi SR, Mumby PJ, Forrest DL & Pinsky ML (2021). Evolution reverses the effect of network structure on metapopulation persistence. *Ecology* 102(7): e03381. <https://doi.org/10.1002/ecy.3381>
4. Tekwa EW, **McManus LC**, Greiner A, Colton MA, Webster MS & Pinsky ML (2021). Geometric Analysis of Regime Shifts in Coral Reef Communities. *Ecosphere* 12(1): e03319.
<https://doi.org/10.1002/ecs2.3319>
3. **McManus LC**, Vasconcelos VV, Levin SA, Thompson DM, Kleypas JA, Castruccio FS, Curchitser EN & Watson JR (2020). Extreme temperature events will drive coral decline in the Coral Triangle. *Global Change Biology* 26(4): 2120-2133. <https://doi.org/10.1111/gcb.14972>
2. **McManus LC**, Watson JR, Vasconcelos VV & Levin SA (2019). Stability and recovery of coral-algae systems: the importance of recruitment seasonality and grazing influence. *Theoretical Ecology* 12(1): 61-72. <https://doi.org/10.1007/s12080-018-0388-x>
1. **McManus LC**, Yurek S, Teare PB, Dolan TE & Serafy JE. (2014). Killifish habitat suitability as a measure of coastal restoration performance: integrating field data, behavioral trials and simulation. *Ecological Indicators* 44: 173-181. <https://doi.org/10.1016/j.ecolind.2014.03.006>

OTHER PUBLICATIONS

Vasconcelos VV*, Marquitti F*, Ong T*, **McManus LC***, Aguiar M, Campos AB, Dutta PS, Jovanelly Km Junquera V, Kong J, Krueger EH, Levin SA, Liao W, Lu M, Mittal D, Pascual M, Pinheiro FL, Rocha J, Santos FP, Sloom P, Su C, Taylor B, Tekwa E, Terpstra S, Tilman AR, Watson JR, Yang L, Yitbarek S & Zhan Q. (2023). Rate-Induced Transitions in Networked Complex Adaptive Systems: Exploring Dynamics and Management Implications Across Ecological, Social, and Socioecological Systems. arXiv preprint: <https://doi.org/10.48550/arXiv.2309.07449>

McClanahan TR, Darling ES, Oddenyo R, Surya G, Beger M, Fox F, Jupiter SD, McLeod L **McManus LC**, van Woesik R, Grantham H, Logan C, Maina J, Patankar V, Wenger A & Zinke J (2022). Forecasting

McManus JW & **McManus LC** (2012). Proposed Dredging for an Aircraft Carrier Turning Basin in Apra Harbor, Guam: Options for Assessment and Mitigation. Technical Report. Engineer Research and Development Center, U.S. Army Corps of Engineers. 121 pages.

GRANTS

\$6,662,478 in total funding; \$5,905,148 to the University of Hawai'i at Mānoa

- | | |
|-----------|--|
| 2025-2030 | NSF Biological Oceanography & NSF Mathematical Biology, <i>CAREER: An integrative framework for modeling coral adaptive potential under climate change</i> . McManus LC . (\$999,997 to UHM) |
| 2025-2028 | NSF Biological Oceanography, <i>Coral reef responses to climate-induced wildfire disturbance</i> . Wall CB, McManus LC & Van Heukelem L. (\$850,000 to UHM) |
| 2024-2026 | Hawai'i Sea Grant College Program, <i>A framework to elucidate historical dynamics and support adaptive management on Hawaiian coral reefs</i> . McManus LC , Okano R, Rahnke S, Rii S & Winter K. (\$93,682 to UHM) |
| 2024-2026 | Hawai'i Sea Grant College Program, <i>Effects of loko i'a (fishpond) restoration on climate-dependent ecosystem dynamics in Kāne'ohe Bay, O'ahu, Hawai'i</i> . McManus LC , Innes-Gold AA, Madin EMP, Rii Y & Winter K. (\$99,261 to UHM) |
| 2024-2026 | NOAA Office of Marine Sanctuaries, <i>Eco-evolutionary modeling to inform coral reef management within National Marine Sanctuaries</i> . Toonen RJ & McManus LC . (\$257,263 to UHM) |
| 2023-2027 | NOAA Ocean Acidification Program, <i>mCDR 2023: Assessing the effects and risks of ocean alkalinity enhancement on the physiology, functionality, calcification, and mineralogy of corals and crustose coralline algae in the Pacific</i> . Melendez M, Bahr K, Sabine CL, Hawco NJ, McNicholl C, McManus LC & Barkley CH. (\$1,999,835 to UHM) |
| 2023-2026 | NSF Infrastructure Innovation for Biological Research, <i>Model enabled machine learning (MnML) for predicting ecosystem regime shifts</i> . Watson JR & McManus LC . (\$957,674 total; \$199,984 to UHM) |
| 2023-2026 | NSF Biological Oceanography, <i>Predicting spatial and temporal responses of herbivorous coral reef fishes to sediment runoff</i> . Johansen JL, McManus LC & Rivera MJ. (\$1,173,835 to UHM). |
| 2022-2024 | Hawai'i Sea Grant College Program, <i>Working towards sustainability of Hawai'i's nearshore fisheries through characterizing and modeling fisheries regulation effects</i> . Madin EMP, Oleson KLL, McManus LC , Innes-Gold AA & Rago Z. (\$231,291 to UHM). |

FELLOWSHIPS & AWARDS

2025	Ecological Society of America Early Career Fellow
2015	Princeton EEB Women Scientists in Conservation Biology Research Award
2015	Best Poster: NMFS-Sea Grant Fellowship Symposium
2014-2017	NMFS-Sea Grant Fellowship in Population and Ecosystem Dynamics
2014-2017	National Defense Science and Engineering Graduate Fellowship
2014	Princeton Environmental Institute Walbridge Fund Graduate Award

PRESENTATIONS

Invited

2025	Know Your Ocean Speaker Series, Maui Nui Marine Resource Council, virtual. Early Warning Systems for Coral Reefs: Machine Learning for Conservation.
2025	Ecology & Evolution Seminar Series, University of Chicago, Chicago, IL. Eco-evolutionary insights and machine learning to advance coral reef science.
2024	Biodiversity Research Seminar Series, University of British Columbia, Vancouver, Canada. Eco-evolutionary insights and machine learning to advance coral reef science.
2024	NOAA Climate and Global Change Postdoctoral Fellowship Summer Institute, Steamboat, CO. Coral adaptation under climate change: challenges and opportunities.
2023	Serrapilheira/ICTP-SAIFR, São Paulo, Brazil. Adaptation in coral metapopulations.
2021	Marine Science Seminar, Hawai'i Pacific University, Kāne'ohe, HI. Adaptive potential, spatial management, and pattern formation on coral reef systems.
2021	Special Seminar on Theoretical Ecology, International Forum on Advanced Environmental Sciences and Technology, University of Oklahoma, virtual. Conservation for evolving coral populations.
2021	NOAA Marine Protected Areas Center Webinar, virtual. Coral reef eco-evolutionary dynamics: Adaptation and connectivity in MPA networks under future climate change.
2021	Fisheries Ecology Division, NOAA Southwest Fisheries Science Center, virtual. Conservation for evolving coral populations.
2021	Marine Biology Graduate Program Seminar, University of Hawai'i at Mānoa, virtual. Coral reef eco-evolutionary dynamics: theory and conservation applications.

- 2020 School of Aquatic and Fisheries Sciences, University of Washington, virtual. Coral reef eco-evolutionary dynamics: theory and conservation applications.
- 2019 Smithsonian Environmental Research Center, Edgewater, MD. Coral reef dynamics in a changing world: a multiscale perspective.
- 2019 Department of Ecology, Evolution and Natural Resources, Rutgers University, New Brunswick, NJ. Coral reef dynamics in a changing world: a multiscale perspective.
- 2018 Ecological Society of America Annual Meeting, New Orleans, LA. Spatial marine metacommunity connectivity and the response of the Coral Triangle to climate change.
- 2015 NOAA Pacific Islands Fisheries Science Center, Honolulu, HI. Linking dispersal scales, genetic differentiation and persistence in corals.

Contributed

- 2023 Ecological Society of America Annual Meeting, Portland, OR. Eco-evolutionary dynamics in large marine ecosystems under climate warming.
- 2022 International Coral Reef Symposium, Bremen, Germany. Conservation for evolving coral populations.
- 2021 International Coral Reef Symposium, virtual. Network structure drives the eco-evolutionary response of coral populations to climate change.
- 2019 Ecological Society of America Annual Meeting, Louisville, KY. Dispersal network structure constrains eco-evolutionary response under directed environmental change.
- 2018 Ocean Sciences Meeting, Portland, Oregon. Ecological implications of thermal stress and larval connectivity in the Coral Triangle.
- 2016 International Coral Reef Symposium, Honolulu, HI. Larval dispersal as a mechanism for coral persistence on reef communities.
- 2015 Ecological Society of America Annual Meeting, Baltimore, MD. Larval dispersal as a mechanism for coral persistence on reef metacommunities.

WORKSHOPS

-
- 2024 Participant, NOAA Climate and Global Change Postdoctoral Fellowship Summer Institute, Steamboat Springs, CO. Organizers: Wolding BO and Keys PW.
 - 2023, 2024 Organizer, Model-Enabled Machine Learning Meeting, HIMB, Kāneʻohe HI.
 - 2022 Participant, ARC Centre of Excellence for Coral Reef Studies Adaptive Seascapes Workshop, Townsville, Australia. Organizer: Harrison HB.

- 2022 Co-organizer, Banff International Research Station for Mathematical Innovation and Discovery Workshop: Rate-induced transitions in networked systems, Banff, Canada. Organizers: Vasconcelos VV, Darcie Marquitti FM, [McManus LC](#) & Ong TW.
- 2021 Participant, Vibrant Oceans Initiative and Wildlife Conservation Society 50 Reefs Workshop, virtual. Organizers: Darling ES & McClanahan TR.

II. TEACHING & MENTORSHIP

TEACHING

Courses

- | | |
|------------------------|---|
| 2021, 2023, 2024, 2025 | MBIO 610 Mathematical Ecology of Marine Systems, UH Mānoa (Lead Instructor; 3-credit intensive) |
| 2025 | MBIO 691J Global Climate Data: Past, Present and Future, UH Mānoa (Co-instructor; 1-credit intensive) |
| 2024 | MBIO 691I Introduction to Scientific Computing, UH Mānoa (Co-instructor; 1-credit intensive) |

Workshops

- | | |
|------|--|
| 2025 | Modeling time series with scientific machine learning, Ecological Society of America Annual Meeting, Baltimore, MD (Co-organizer; ~65 participants) |
| 2025 | Machine Learning for Understanding Changing Ecosystems, UH Mānoa (Co-organizer; 5-day workshop; 10 undergraduate and graduate student participants) |
| 2025 | Coding the Coast: Marine Systems Modeling in Python, UH Mānoa & Windward Community College (Co-organizer; 3-day workshop; 10 undergraduate student and post-baccalaureate participants) |
| 2024 | Theory of Microbial Symbiosis Workshop, UH Mānoa (Co-organizer; 3-week workshop; 12 graduate student and postdoctoral researcher participants) |
| 2023 | Metapopulation Dynamics, Serrapilheira/ICTP-SAIFR Training Program in Quantitative Ecology, São Paulo, Brazil (Lead Instructor; 1-week workshop; 30 undergraduate student, post-baccalaureate and graduate student participants) |

2023, 2024 Time Management Workshop, Moku o Lo'e Student Association, UH Mānoa (Lead Instructor; 1-day workshop; 20 participants)

Guest lectures

2021, 2022, 2023 Corals and Coral Reefs (BIOL 411), Coral Adaptive Capacity. UH Mānoa

2018 Graduate Seminar: Ecology, Evolution, and Climate Change. Rutgers University

2013 Marine Biology (EEB 312), Connectivity of Coral Reef Ecosystems. Princeton University

STUDENT ADVISING, COMMITTEES, & MENTORING

Primary Advising

Advisee	Position	Institution
Noam Vogt-Vincent	Postdoc (2023-2025)	UHM
Mariana Rocha de Souza (co-advised with Robert Toonen, UHM)	Postdoc (2023-present)	UHM
David Anderson	Postdoc (2024-present)	UHM
Anne Innes-Gold (co-advised with Elizabeth Madin, UHM)	PhD candidate (2022-2025)	UHM
Sophia Rahnke	PhD candidate (2021-present)	UHM
Nathan Fitzpatrick	PhD student (2023-present)	UHM
Jacob Snyder	PhD student (2023-present)	UHM
Shelbie Ishimaru	PhD student (2024-present)	UHM
Amanda Campos	Visiting PhD student (2024-2025)	Universidade de São Paulo, Brazil
Seychelle Brainard	Undergraduate intern (2024-present)	Smith College, MA
Joshua Brett Pablo	Undergraduate intern (2023-2025) Lab technician (2025-present)	UHM
Sophia Girand	Undergraduate intern (2023)	Princeton University, NJ
Eojin Lee	Undergraduate intern (2021-2024)	Columbia University, NY
Katherine Lew	High school intern (2020)	Millburn High School, NJ

Graduate Committees

Advisee	Degree	Institution and Program	Primary Advisor
Toby Matthews	PhD (2022-2024)	UHM MBGP	Megan Donahue, HIMB, UHM
Philip Patton	PhD (2022-2025)	UHM MBGP	Lars Bejder, HIMB, UHM
Nakoa Goo	PhD (2020-present)	UHM MBGP	Robert Toonen, HIMB, UHM
Leon Tran	PhD (2020-present)	UHM MBGP	Jacob Johansen, HIMB, UHM

Devynn Wulstein	PhD (2020-present)	UHM MBGP	Joshua Madin, HIMB, UHM
Cameron Walsh	PhD (2021-present)	UHM MBGP	Brian Bowen, HIMB, UHM
Paige Wernli	PhD (2021-present)	UHM MBGP	Carl Meyer, HIMB, UHM
Samantha Shedd	PhD (2022-present)	UHM MBGP	Cindy Hunter, UHM
Guan-Yan Chen	PhD (2023-present)	UHM MBGP	Joshua Madin, HIMB, UHM
Jon Ehrenberg	PhD (2023-present)	UHM MBGP	Joshua Madin, HIMB, UHM
Kerri Luttrell	PhD (2023-present)	UHM MBGP	Michael Rappé, HIMB, UHM
Danielle Barnas	PhD (2023-present)	UHM MBGP	Megan Donahue, HIMB, UHM
Jasmine Reighard	PhD (2024-present)	UHM MBGP	Jacob Johansen, HIMB, UHM
Maryann Webb	PhD (2024-present)	UHM MBGP	Robert Toonen, HIMB, UHM
Hayley Bedwell	PhD (2021-present)	University of Texas at Austin - Ecology, Evolution and Behavior	Mikhail Matz (University of Texas at Austin)
Jaelyn Bos	PhD (2023-present)	University of California, Santa Cruz - Ecology and Evolutionary Biology	Malin Pinsky (University of California, Santa Cruz)

External Examiner for International PhD Students

Liam Lachs, Newcastle University, 2024

Kevin Bairos-Novak, James Cook University, 2023

III. SERVICE & OUTREACH

PROFESSIONAL SERVICE

University Service

2025 Search Committee Co-chair, HIMB Marine Megafauna Physiologist

2025-present Member, Departmental Personnel Committee, HIMB

2025-present Member, Planning Committee, HIMB

2024-present Chair, Visitor Committee, HIMB

2023-2024 Working Group Leader, Community and Collaboration, HIMB Strategic Planning Initiative

2023 Participant, HIMB Malama 'Aina Climate Change Adaptation Initiative Workshop

2022-2023 Search Committee Chair, HIMB Community Education Coordinator

2022-present Faculty Sponsor, HIMB Moku o Lo'e Student Association

2022, 2023 Member, Marine Biology Graduate Program Admissions Committee

2021 Member, Search Committee, HIMB Biocultural Ecologist

2020-2025 Member, HIMB Departmental Personnel Committee

Professional Service

2025 Lead organizer, Modeling Coral Reef Ecosystems Across Scales Symposium, International Coral Reef Symposium, Auckland, New Zealand (2026)

2023 Steering committee member for the Ecological Society of America Marine and Coastal Ecology Section

2022 Co-organizer, Can evolution rescue corals from the effects of climate change? Symposium, International Coral Reef Symposium, Bremen, Germany

2021, 2022 National Science Foundation review panels

Peer Review

Journals: American Naturalist, Climatic Change, Climate Resilience and Sustainability, Coral Reefs, Functional Ecology, Ecology, Global Change Biology, Global Ecology and Biogeography, Journal of Applied Ecology, Nature Climate Change, Theoretical Ecology

CONSERVATION & MANAGEMENT

2023- present Collaboration with the NOAA Office of Marine Sanctuaries to develop a modeling tool that can compare the effectiveness of management strategies on reefs in the proposed Pacific Remote Islands Marine National Monument.

2022- 2025 Collaboration with the State of Hawai'i Division of Aquatic Resources, Native Hawaiian natural resource managers, and local community members from Kāne'ohe, Oahu to co-develop research that will support the implementation of Indigenous resource management Hawai'i. Recent awarded Hawai'i Sea Grant proposals were co-developed with stakeholders.

2021 Vibrant Oceans Initiative and Wildlife Conservation Society 50 Reefs Workshop Participant. Co-authored a white paper to aid coral reef stakeholders in prioritizing sites for management and protection.

McClanahan et al. (2022). Forecasting Climate Sanctuaries for Securing the Future of Coral Reefs. <https://wcs.org/coral-science-whitepaper>

2017-present Ongoing collaboration with the Coral Reef Alliance, an NGO focused exclusively on coral reef protection, to disseminate marine protected area policy recommendations based on eco-evolutionary coral reef research.

Informational pamphlet. Evolution can rescue coral reefs—and you can help. (2022). Developed by Coral Reef Alliance in consultation with [McManus LC](#).

OUTREACH & SCIENCE COMMUNICATION

2025	Interactive ecological modeling booth, HIMB Open House, Kāneʻohe, HI
2024	Science and painting activity of 3D-printed Heʻeia ahupuaʻa models, KEY Project, Kāneʻohe, HI
2024	Interactive ecological modeling booth, Mauka to Makai Earth Day Event at Waikīkī Aquarium, Honolulu, HI
2023-2025	Advisor, Symphony of the Hawaiʻi Seas, Hawaiʻi Sea Grant and Hawaiʻi Symphony Orchestra
2022	Co-organizer, Communication Creativity Class, UH Mānoa Communications School and Hawaiʻi Institute of Marine Biology. (Instructor: Amaral, PAK).
2018, 2019	New Brunswick High School Build It Better Design Challenge, New Brunswick, NJ
2018	Instructor, Agent-based modeling with NetLogo, New Brunswick High School STEM Club, New Brunswick, NJ
2018	Mentor, Rutgers Future Scholars High School Internship Program, Rutgers University

SELECTED MEDIA COVERAGE

2025	“HIMB Researcher Lisa McManus Awarded 2025 Ecological Society of America Early Career Fellowship”; UH Mānoa press release
2024	“Could fishponds help with Hawaiʻi’s food sustainability?”; Various media outlets including Mongabay based on a UH Mānoa press release
2022	“Scientists Believe Evolution Could Save Coral Reefs, If We Let It”; Various media outlets including Science Daily based on a Rutgers University press release
2022	“Cultivating Super Corals Alone is Unlikely to Protect Coral Reefs from Climate Change”; Various media outlets including Phys.org based on a Rutgers University press release
2022	“Large-scale ocean sanctuaries could protect reefs from climate change”: Various media outlets including Science Daily based on a Ohio State University press release